

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (EE) Examination

December 2012

EE703 Automation & Control System

Date: 20.12.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

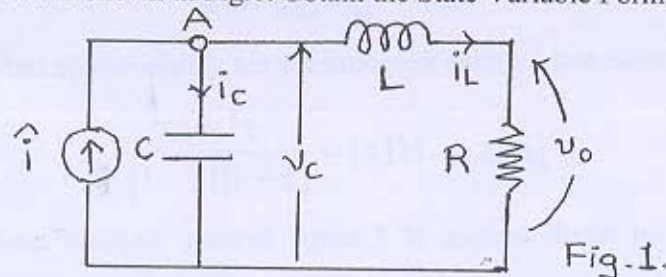
Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of the scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain State Transition Matrix for State Variable Formulation. [04]
(b) Consider network shown in Fig.1. Obtain the State Variable Formulation. [03]



- Q - 2 (a) Describe Controllability and Observability for Modern Control System. [07]
(b) Explain Design of State Observer with necessary diagram. Specify the Condition for the existence of observer. [07]

OR

- Q - 2 (a) Describe important properties of Eigen values of a Matrix for State Variable Formulation. Give explanation of Duality Property of multi-input-multi-output linear time invariant system. [07]
(b) Specify all the Methods of State Space Representation in Canonical Form. Explain any two of them in detail. [07]

- Q - 3 (a) Compare Digital Controller and Analog controller of Control System. [03]
(b) Determine Z-transform and sketch the ROC(Region of Convergence) of the sequence. [04]

$$x[n] = [5(9^n) - 12(8^n)]u(n)$$

- (c) Elucidate relationship between S-plane and Z-plane poles. [07]

OR

- Q - 3 (a) Define Sampling Process. Explain Sample and Hold Device in with necessary diagrams. [03]

- (b) Prove Differentiation in Z domain property of Z-Transform. By using this property, [04]
Determine Z-Transform and sketch ROC of given sequence.

$$x[n] = na^n u(n)$$

- (c) Find Inverse Z-Transform of given $X[z]$ using Partial Fraction Expansion Method. [07]

$$X[z] = \frac{z^2}{[z-0.5][z-1]^2}$$

SECTION - II

- Q - 4 (a) Find Inverse Z-Transform of given $X[z]$ using Residue Method [04]

$$X[z] = \frac{1}{[z-1][z-0.5]}$$

- (b) Find Step Response and Impulse Response for the system whose transfer function is [03]

$$\text{given as } H[z] = \frac{z^3 + 1}{[z+1][z-1]}$$

- Q - 5 (a) Classify Common Nonlinearities of Control System. Explain each of them with [07]
necessary figures.

- (b) Define Phase Plane Method. Explain Basic Concept of Phase Plane Method with [07]
necessary example.

OR

- Q - 5 (a) Explain Asymptotic Stability Analysis with necessary example. [07]

- (b) Illustrate Stability, Instability and Sign Definiteness for Liapunov Stability Analysis. [07]

- Q - 6 (a) Describe the Role of Arm and Wrist of a Robotic Manipulator. [04]

- (b) Specify the Basic Laws of Robotics. [03]

- (c) Explain the Four basic configurations of Arm in Robotic Manipulators. [07]

OR

- Q - 6 (a) Explain Mapping between rotated frames in Robotics. [04]

- (b) Define the following terms for Robotics (i) Load Carrying Capacity(ii) Work Volume [03]
(iii) End-Effector

- (c) Explain Fundamental rotation matrices in robotics. Specify Different Types of [07]
Applications of Robotics.
